MIL-C-45953(WC)

22 January 1974
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

CONTROL PANEL, PILOT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pilot's control used to operate the M28A1 Helicopter Armament Subsystem. The control incorporates switches, panel lights and instruments to indicate weapons employed, operational (circuit) functions and control of the armament subsystem by the pilot.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystem, General Specification For.
MIL-P-14232	-	Parts, Equipment and Tools For Army Materiel, Packaging and Packing of.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

FSC 1010

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DRAWINGS

U.S. Army Weapons Command
D11687987 - Control Panel, Pilot.
IEL11687987 - List of Inspection Equipment.

(Product drawings referenced in this specification form a part of D11687987; acceptance inspection equipment drawings form a part of IEL11687987.)

PUBLICATIONS

U.S. Army Weapons Command
P11687987 - Packaging Data Sheet for Control Panel, Pilot.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include a pilot pack (see 5.1).

3.2 Materials, construction and design. The control panel shall conform to the materials, construction and design requirements specified herein, on Drawing D11687987 and in MIL-W-13855.

3.2.1 Weight. The weight of the control panel shall not exceed 1.3 pounds.

3.3 Functional requirements.

3.3.1 Power inputs. The control panel shall meet the requirements specified herein when supplied with 24 to 30 volts direct current (vdc) with a minimum current of 3.0 ampheres.

3.3.2 Panel edge lights (DS3, DS4). Panel edge lights DS3 and DS4 shall illuminate when 24 - 30 vdc is applied between pin T (positive) and pin V of connector A9J1.

3.3.3 Switches.

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3.3.3.1 Master arm switch (S1), and Gunner/Pilot switch (S2). The master switch shall provide connections between pins A and C of connector A9J1 and the following A9J1 terminals:

Switch (S1) Position	A9J1 Terminals pin B	pin L	pin N	pin H
OFF	Open	Open	Open	Open
SAFE	Continuity	Open	(See Note 1)	(See Note 1)
ARMED	Continuity	Continuity	(See Note 1)	(See Note 1)

Note 1: The reading at nonnector A9J1 shall be as follows for switch positions S2:

Switch (S2) Position	A9J1 Terminals pin N	pin H
GUNNER	Continuity	Open
PILOT	Open	Continuity

An open circuit between pin X and pin Y of connector A9J1 shall exist when switch (S1) is in the OFF or SAFE position. Continuity shall exist between pin X and pin Y when switch (S1) is in the ARMED position.

3.3.3.2 Weapon select switch (S3). The weapon select switch shall provide the following connections to A9J1 terminals.

Switch (S3) Position	A9J1 Terminals pins U and S	pins U and R
BOTH	Connected	Connected
LEFT	Open Circuit	Connected
RIGHT	Connected	Open Circuit

3.3.3.3 POINT/AREA fire switch (S4). With the fire switch in the POINT position, an open circuit shall exist between pin W and pin M of connector A9J1. Continuity shall exist between pin W and pin M when the fire switch is in the AREA position.

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3.3.4 Indicators.

3.3.4.1 Indicator lights. Indicator lights with individual dimming adjustments and press-to-test switch shall illuminate when 24-30 vdc is applied between pin V (GND) and pins A and C of connector A9J1.

3.3.4.2 Armed power indicator (DS1). The amber light shall illuminate when 24 - 30 vdc is applied between pin D (positive) and pin V of connector A9J1.

3.3.4.3 Turret power (DS2). The green light shall illuminate when 24 - 30 vdc is applied between pin G (positive) and pin V of connector A9J1.

3.3.4.4 7.62mm, left (DS5). The blue light shall illuminate when 24 - 30 vdc is applied between pin P (positive) and pin V of connector A9J1.

3.3.4.5 40mm, left (DS6). The blue light shall illuminate when 24 - 30 vdc is applied between pin F (positive) and pin V of connector A9J1.

3.3.4.6 7.62mm, right (DS7). The blue light shall illuminate when 24 - 30 vdc is applied between pin J (positive) and pin V of connector A9J1.

3.3.4.7 40mm, right (DS8). The blue light shall illuminate when 24 - 30 vdc is applied between pin E (positive) and pin V of connector A9J1.

3.4 Marking. The control panel shall be marked in accordance with MIL-W-13855.

3.5 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s) shall be submitted for inspection in accordance with contract requirements (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection lots. The formation, size, and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Control panels shall be assembled from component parts that have met all inspection requirements specified herein.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the criteria specified in the contract (see 6.2). Examination of these parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.5.2.2 Control panels. Each control panel shall be examined as specified in Table I. The examination provisions should be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspection of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. Control panels failing to meet these requirements shall be rejected.

4.5.2.2.1 Inspection methods. The following provisions shall be applicable to the prescribed inspection methods. Request for a method other than that specified shall be submitted to the Government for approval.

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a. Where "V$sual" is specified as the inspection method for assemblies, the assembly shall be visually examined for completeness, security, function and conformance to specified requirements.

TABLE I

Characteristic	Requirement	Inspection Method
a. Missing, damaged or incorrectly assembled component parts.	3.2	Visual
b. Illegible, incorrect or missing product markings and identification plates.	3.4	Visual
c. Workmanship.	3.5	Visual

4.5.2.2.2 Weight. The first 3 control panels produced shall be examined for the weight requirement specified in 3.2.1.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete control panel. If test requirements cited herein are not met, acceptance of control panels shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional. physical, and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defects in future production.
c. Examine control panels, partially assembled control panels, and components (including components and sub-assemblies at in-process or final assembly) to assure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned control panel for retest.

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4.5.3.2 Functional testing. Each control panel shall be function tested (see 3.3) using the test method specified in 4.6.1. Control panels failing to meet functional requirements shall be rejected.

4.5.3.3 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specffied (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Acquisition, maintenance and disposition. Unless otherwise specified, responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment, Drawing IEL11687987 and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

4.6.1 Functional test. Control panels shall be tested in accordance with the operating procedure specified in IEL11687087 (ref Drawing 11836853).

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack shall consist of a complete control panel packaged in accordance with Packaging Data Sheet P11687987 for the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing and marking. Control panels shall be preserved, packaged, packed and marked in accordance with

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Packaging Data Sheet P11687987 for the level of protection specified in the contract (see 6.2).

5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or the requirements specified, and for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The pilot control is intended for use on, but not limited to the M28A1 Helicopter Armament Subsystem.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Lists of drawings and specifications pertinent to the control panel showing applicable revision dates.
c. Shipping instructions for first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Criteria for examination and testing of component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.3).
e. Packaging examination and testing, if different (see 4.5.4).
f. Responsibilities for acquisition, maintenance and disposition of acceptance inspection equipment if other than specified (see 4.5.5.1).
g. Selection of applicable levels of preservation, packaging and packing (see 5.2).
h. Packaging instruction for repair parts (see 5.3).

6.2.2 Contract data requirements. When warranted, the requirements for records of examinations and tests and failure analysis data will be listed on a DD Form 1423 incorporated into the contract.

6.2.3 Management control systems requirement. When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.2.4 Unless otherwise specified (see 6.1.1 f), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

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6.3 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description, USAWECOM PD-29, Amendment 2, dated 31 March 1972.

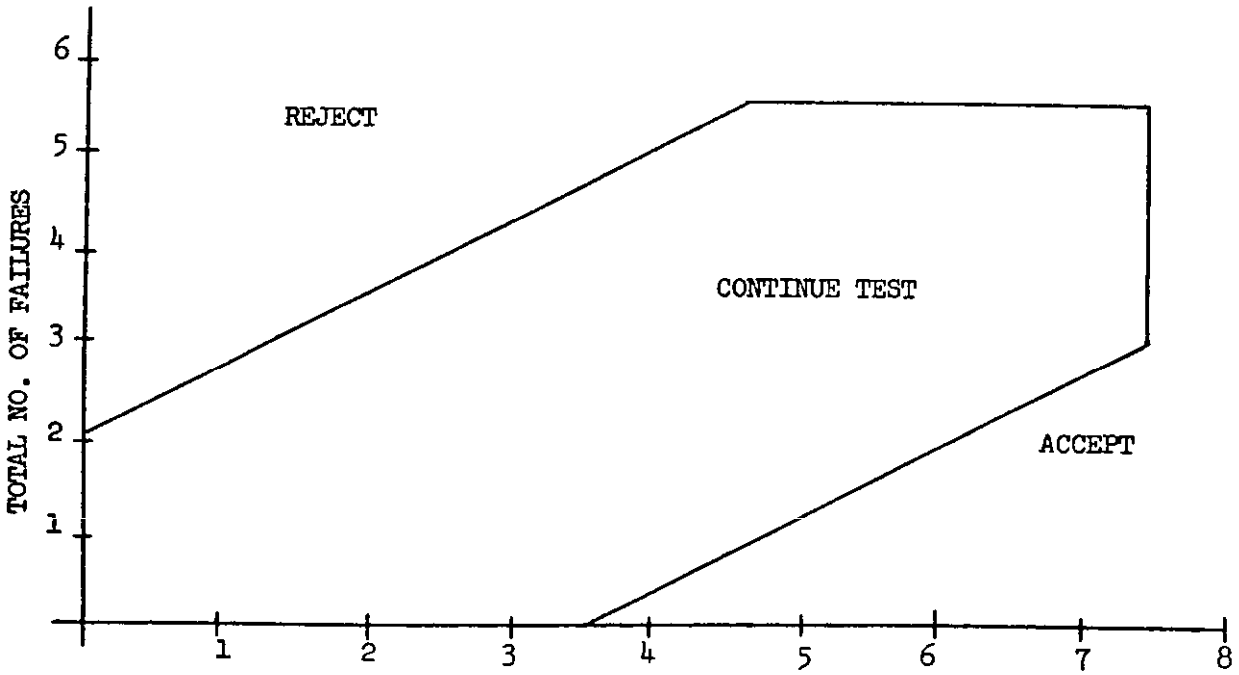
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M28A1 SUBSYSTEM:
PILOT CONTROL PANEL
Decision Risks: 30%
Discrimination Ratio: 1.5:1



Test Cycles X 1000

NO. OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	3418
1	N/A	4736
2	N/A	6054
3	1318	7373
4	2637	7373
5	3955	7373
6	7373	N/A

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-C-45953(WC): Control Panel, Pilot

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. *IS THE DOCUMENT RESTRICTIVE?*
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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